

Final Analysis: Somatic Polygenic Risk Scoring

University of Michigan BIOSSTAT 629 Winter 2026

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This quarto document contains all the code to generate all tables, figures, and results from Somatic Polygenic Risk Scoring: A Novel Framework for Metastasis Risk Stratification Using Somatic Genetic Profiles.

Sections

0. Reproducibility settings, loading libraries and data, data pre-processing
1. Cohort generation (Data Splits)
2. Lasso regression
3. Lasso regression evaluation via bootstrapping
4. sPRS computation
5. sPRS evaluation
6. sPRS performs better than standard metastasis measures?
7. Final tables and figures

Section 0: Reproducibility settings, libraries and data, and pre-processing

Mutations Data Pre-Processing

The first step is to select the genes that are common across all panels. This is to reduce misrepresentation of mutation/ wildtype indicators in the mutations that are being considered.

The mutations data set needs to be pivoted wider to create a data frame where each row is a single unique patient, and the rest of the columns are indicators of specific mutation/variant type combinations.

Then, need to select which variant classification system we are using (simple INS, DEL, SNP, DNP, etc OR more detailed non-sense mutation, missense mutation, in frame deletion, frameshift

deletion, etc.). To do so, create two data frames with a mutation x variant type indicator column for each patient and see what proportion of patients have each type of classification by gene. If there are very small proportions, use a collapsed variant classification system.

The side-by-side comparison of raw proportions of variations across the whole data set below shows that there are not super drastic differences in proportions between using variant class or variant type. Therefore, proceed using variant classification as that provides more information about the genetic mutation itself, and reserve variant type as a secondary analysis.

	Gene_VarClass	Proportion	Gene_VarType	Proportion
1	TP53_Missense_Mutation	0.36264252	TP53_SNP	0.46073366
2	KRAS_Missense_Mutation	0.30272879	KRAS_SNP	0.29653670
3	PIK3CA_Missense_Mutation	0.15117223	PIK3CA_SNP	0.15074519
4	APC_Nonsense_Mutation	0.12298757	APC_SNP	0.14344280
5	TP53_Nonsense_Mutation	0.07947218	TP53_DEL	0.07814835
6	APC_Frame_Shift_Del	0.07635478	APC_DEL	0.07750779
7	TP53_Frame_Shift_Del	0.05867532	SMAD4_SNP	0.06021267
8	EGFR_Missense_Mutation	0.05363625	EGFR_SNP	0.05286757
9	BRAF_Missense_Mutation	0.04577871	ATM_SNP	0.04979289
10	SMAD4_Missense_Mutation	0.04535167	KMT2D_SNP	0.04889610
11	FAT1_Missense_Mutation	0.04287483	FAT1_SNP	0.04757228
12	KMT2D_Missense_Mutation	0.04159371	BRAF_SNP	0.04577871
13	TP53_Splice_Site	0.04022719	KMT2C_SNP	0.04441218
14	ATM_Missense_Mutation	0.03898877	ARID1A_SNP	0.04317376
15	KEAP1_Missense_Mutation	0.03890336	FBXW7_SNP	0.04257591
16	GRIN2A_Missense_Mutation	0.03582867	KEAP1_SNP	0.04253320
17	EGFR_In_Frame_Del	0.03441944	NF1_SNP	0.04159371
18	FBXW7_Missense_Mutation	0.03330913	SMARCA4_SNP	0.03890336
19	KMT2C_Missense_Mutation	0.03232694	GRIN2A_SNP	0.03719520
20	EPHA5_Missense_Mutation	0.03181449	EGFR_DEL	0.03510270
21	SMARCA4_Missense_Mutation	0.03121664	EPHA5_SNP	0.03441944
22	APC_Frame_Shift_Ins	0.03104582	CDKN2A_SNP	0.03352265
23	NOTCH3_Missense_Mutation	0.03010633	APC_INS	0.03262587
24	ERBB4_Missense_Mutation	0.02955118	ERBB4_SNP	0.03219883
25	PIK3CG_Missense_Mutation	0.02869710	NOTCH3_SNP	0.03147286
26	EPHA3_Missense_Mutation	0.02788572	STK11_SNP	0.03134475
27	ALK_Missense_Mutation	0.02775761	TERT_SNP	0.03087501
28	NOTCH1_Missense_Mutation	0.02762950	ATRX_SNP	0.03053337
29	CTNNB1_Missense_Mutation	0.02733057	PTEN_SNP	0.03049067
30	CREBBP_Missense_Mutation	0.02707435	CREBBP_SNP	0.03049067

Section 1: Cohort Generation

Now, create a final data set with all patient demographic, sample (metastasis information), and mutation information. Duplicated patients are removed, and all missing observations are dropped. This will be used in all following analyses.

	Metastasis	Primary
Breast Cancer	0.10060470	0.11538634
Colorectal Cancer	0.06159015	0.17684211
Non-Small Cell Lung Cancer	0.10898096	0.19520717
Pancreatic Cancer	0.04676372	0.08129899
Prostate Cancer	0.04170213	0.07162374

```
[1] "Data set splits: train and test"
```

```
[1] 0.6999776
```

```
[1] 0.3000224
```

```
[1] "Data split for train by cancer type and metastasis status"
```

	Metastasis	Primary
Breast Cancer	0.10552249	0.13681449
Colorectal Cancer	0.06552761	0.15575606
Non-Small Cell Lung Cancer	0.11467332	0.19850259
Pancreatic Cancer	0.04332245	0.07071095
Prostate Cancer	0.04268254	0.06648749

```
[1] "Data split for test by cancer type and metastasis status"
```

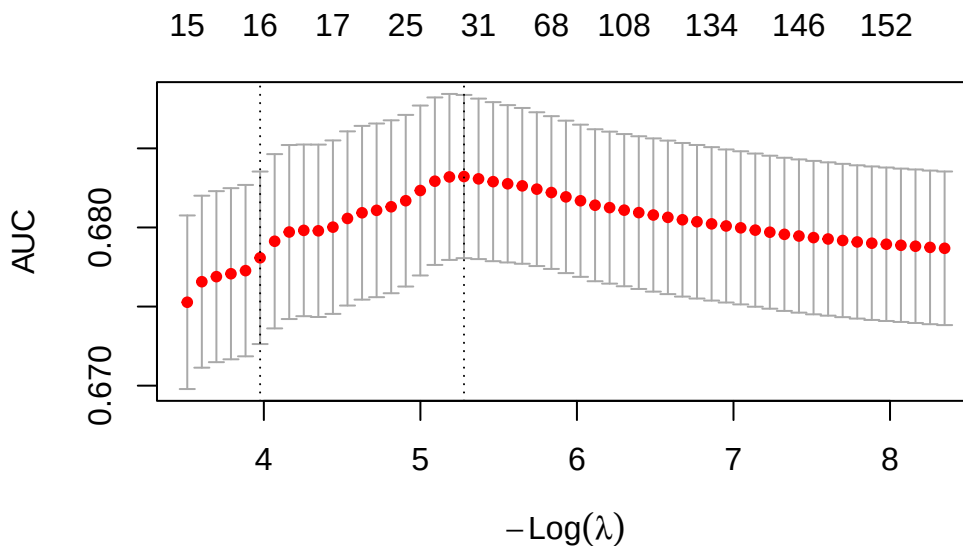
	Metastasis	Primary
Breast Cancer	0.08913108	0.06539265
Colorectal Cancer	0.05240370	0.22603762
Non-Small Cell Lung Cancer	0.09570021	0.18751866
Pancreatic Cancer	0.05479248	0.10600179
Prostate Cancer	0.03941475	0.08360705

Section 2: LASSO regression

Now, fit a LASSO logistic regression that penalizes ONLY the mutation x gene variables. The covariates are NOT penalized. This is to adjust for demographics within the sPRS score, so sPRS score will reflect the broader population (MSK-CHORD patient population). A 10-fold CV among 100 lambdas shows which lambda is the best penalty factor. All done on TRAINING data split.

Section 3. Evaluation of LASSO model via Bootstrapping

Now, identify the best lambda by plotting the auc across the 10 folds, comparing the min lambda to the 1se lambda. Then, the training data set is bootstrapped and the LASSO is fit on those bootstrapped samples to assess the stability of the chosen mutations. Finally, the surviving mutations are identified and will be used to construct sPRS in the next section.



```
[1] "AUC at min lambda"
```

```
[1] 0.6832202
```

```
[1] "AUC at 1SE lambda"
```

```
[1] 0.6780831

[1] "How many gene x mutation type survive at min labda?"

s19
15

[1] "How many gene x mutation type survive at 1SE lambda?"

s5
1

[1] "How many mutations survive bootstrapping?"

[1] 10

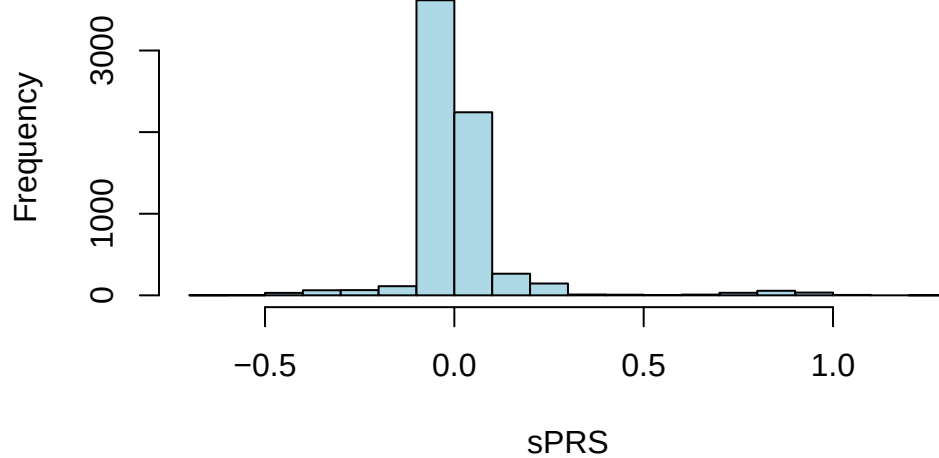
[1] "Betas of surviving stable gene x mutation type"
```

	beta	feature
1	-0.32293056	RNF43_Frame_Shift_Del
2	-0.17121284	FBXW7_Missense_Mutation
3	-0.08295691	KMT2C_Missense_Mutation
4	-0.04003361	PIK3CA_Missense_Mutation
5	0.09496226	TP53_Missense_Mutation
6	0.09898395	AR_Missense_Mutation
7	0.10945141	KEAP1_Missense_Mutation
8	0.11345307	ARID1A_Nonsense_Mutation
9	0.15471594	ARID1B_Missense_Mutation
10	0.83889956	ESR1_Missense_Mutation

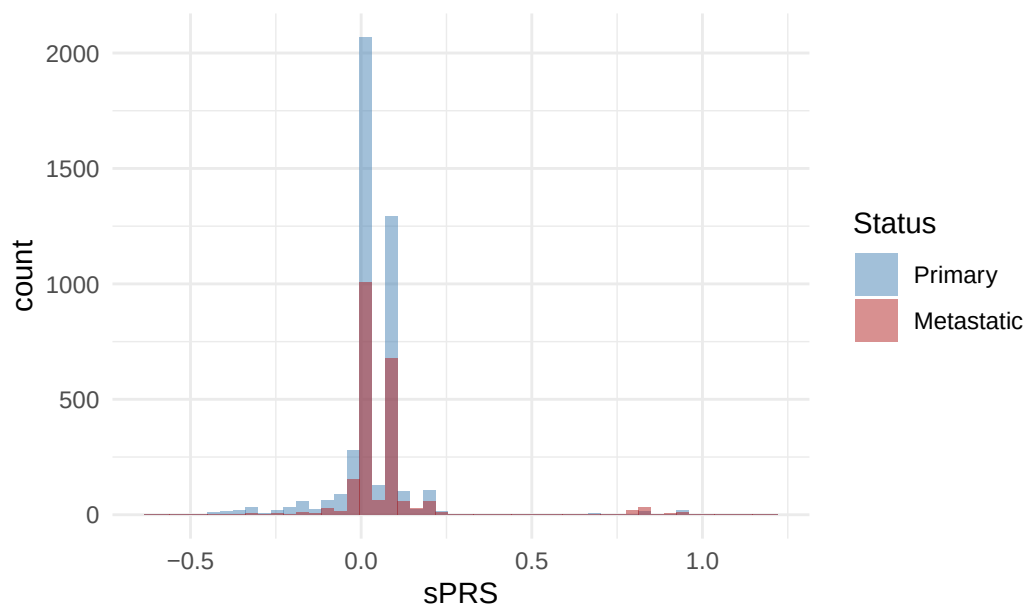
Section 4. sPRS computation

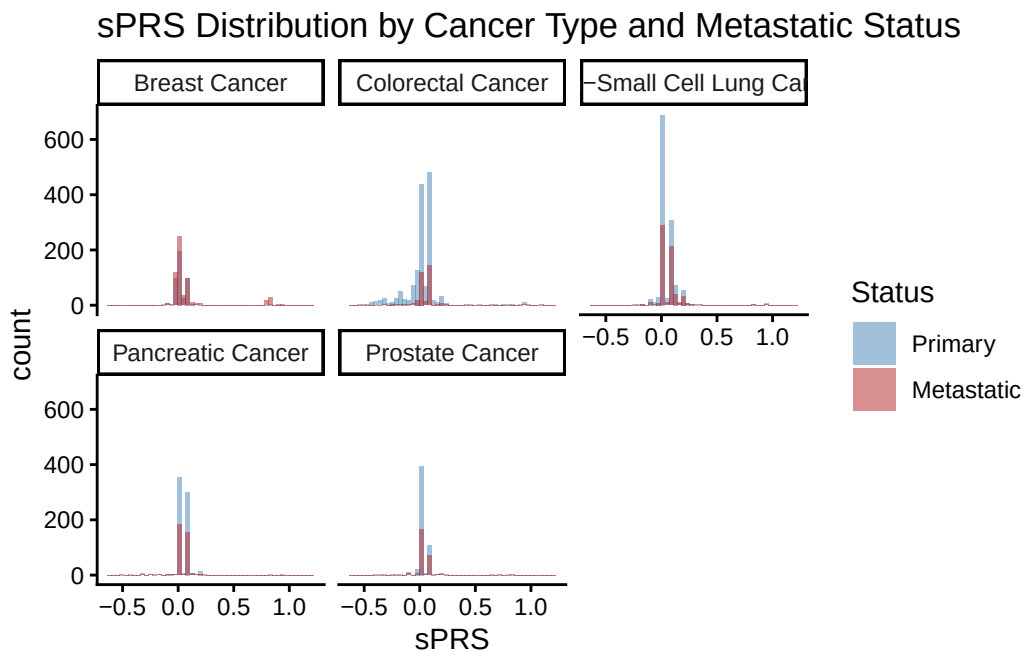
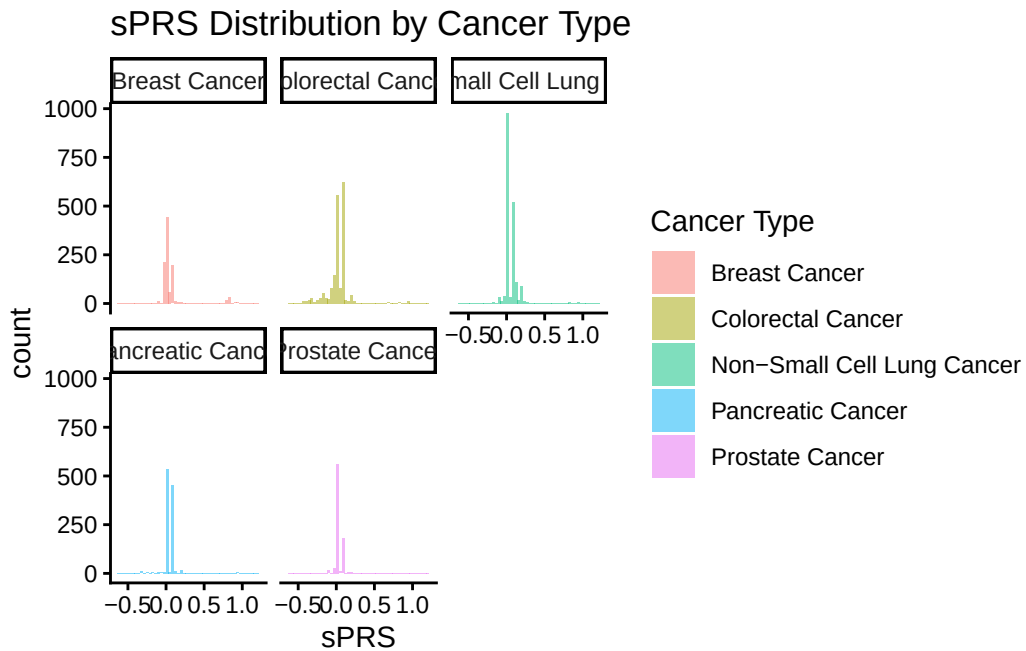
In this section, the weights (betas) from the previous section will be used to create an sPRS score for the test set. This sPRS score will be evaluated by using it as the sole predictor of metastasis.

Overall Distribution of sPRS in Test Set



sPRS Distribution by Metastatic Status





Section 5: sPRS Evaluation

This section fits both an unadjusted and adjusted binomial logistic regression that is answering the following question: is sPRS associated with metastasis status, ignoring all other variables AND is sPRS associated with metastasis status, independent of common cancer demographics?

Unadjusted Evaluation

Classify metastasis status SOLELY with sPRS. This follows the philosophy that the sPRS was already created by aggregating weights that took into account the adjustment of covariates. Thus, no need to adjust AGAIN by covars.

Call:

```
glm(formula = y ~ sPRS, family = "binomial", data = sPRS_test)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.76732	0.02748	-27.925	< 2e-16 ***
sPRS	1.39249	0.17660	7.885	3.14e-15 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 8509.1 on 6697 degrees of freedom
Residual deviance: 8444.7 on 6696 degrees of freedom
AIC: 8448.7

Number of Fisher Scoring iterations: 4

[1] "Odds Ratios"

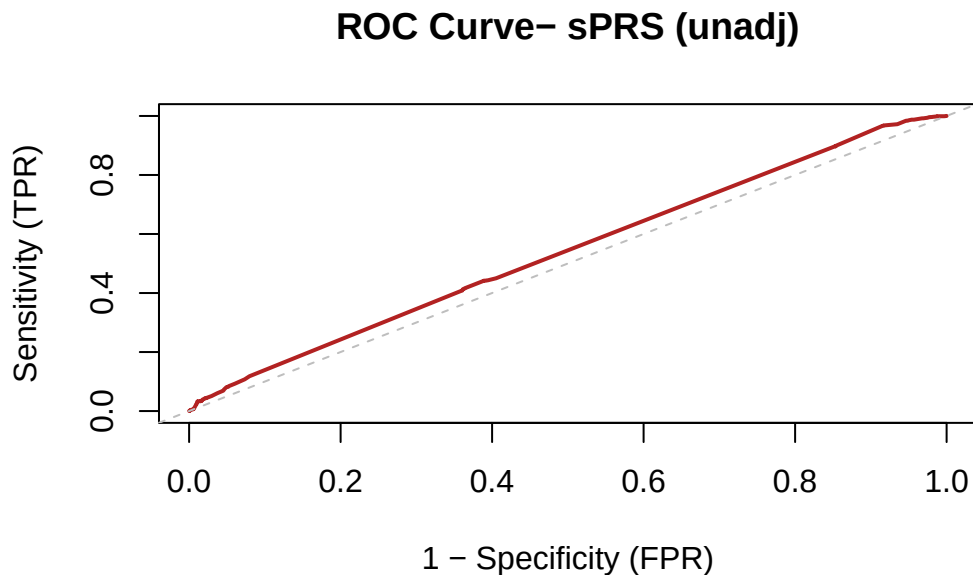
	sPRS
(Intercept)	0.4642576

[1] "95% CIs"

	2.5 %	97.5 %
(Intercept)	0.4398343	0.4898582
sPRS	2.8548825	5.7091260


```
[1] "LRT p-value"
```

```
[1] 1.0354e-15
```



```
[1] "AUC for unadjusted sPRS log reg"
```

```
[1] 0.5416564
```

Adjusted Evaluation

Classify metastasis status with sPRS, adjusting for covars. This follows the philosophy that the sPRS was created with covars adjusted for at the reference groups or at 0, but adjusting specifically for covars when classifying patients' metastasis status will spread signal across what is appropriate and make sPRS fair across demographics.

Call:

```
glm(formula = y ~ sPRS + GENDER + CURRENT_AGE_DEID + as.factor(CANCER_TYPE) +  
     as.factor(RACE), family = "binomial", data = sPRS_test)
```

Coefficients:

	Estimate	Std. Error	z	value
(Intercept)	0.316991	0.169428	1.871	
sPRS	1.274069	0.190840	6.676	
GENDERMale	-0.027418	0.065037	-	
0.422				
CURRENT_AGE_DEID	-0.002154	0.002354	-	
0.915				
as.factor(CANCER_TYPE)Colorectal Cancer	-1.724345	0.093528	-	
18.437				
as.factor(CANCER_TYPE)Non-Small Cell Lung Cancer	-0.928046	0.089156	-	
10.409				
as.factor(CANCER_TYPE)Pancreatic Cancer	-0.901828	0.099148	-	
9.096				
as.factor(CANCER_TYPE)Prostate Cancer	-0.956748	0.120936	-	
7.911				
as.factor(RACE)Black Or African American	0.076559	0.135756	0.564	
as.factor(RACE)Native American-Am Ind/Alaska	-10.483277	135.651752	-	
0.077				
as.factor(RACE)Native Hawaiian Or Pacific Isl	0.959335	0.961980	0.997	
as.factor(RACE)No Value Entered	0.984664	0.967251	1.018	
as.factor(RACE)Other	0.205398	0.178913	1.148	
as.factor(RACE)Pt Refused To Answer	-0.143223	0.210392	-	
0.681				
as.factor(RACE)Unknown	0.085059	0.221500	0.384	
as.factor(RACE)White	0.028195	0.098417	0.286	
	Pr(> z)			
(Intercept)	0.0614	.		
sPRS	2.45e-11	***		
GENDERMale	0.6733			
CURRENT_AGE_DEID	0.3603			
as.factor(CANCER_TYPE)Colorectal Cancer	< 2e-16	***		
as.factor(CANCER_TYPE)Non-Small Cell Lung Cancer	< 2e-16	***		
as.factor(CANCER_TYPE)Pancreatic Cancer	< 2e-16	***		
as.factor(CANCER_TYPE)Prostate Cancer	2.55e-15	***		
as.factor(RACE)Black Or African American	0.5728			
as.factor(RACE)Native American-Am Ind/Alaska	0.9384			
as.factor(RACE)Native Hawaiian Or Pacific Isl	0.3186			
as.factor(RACE)No Value Entered	0.3087			
as.factor(RACE)Other	0.2510			
as.factor(RACE)Pt Refused To Answer	0.4960			
as.factor(RACE)Unknown	0.7010			
as.factor(RACE)White	0.7745			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 8509.1 on 6697 degrees of freedom
Residual deviance: 8003.9 on 6682 degrees of freedom
AIC: 8035.9

Number of Fisher Scoring iterations: 10

[1] "Odds Ratios for intercept and sPRS only (adjusted model)"

(Intercept)	sPRS
1.37299	3.57537

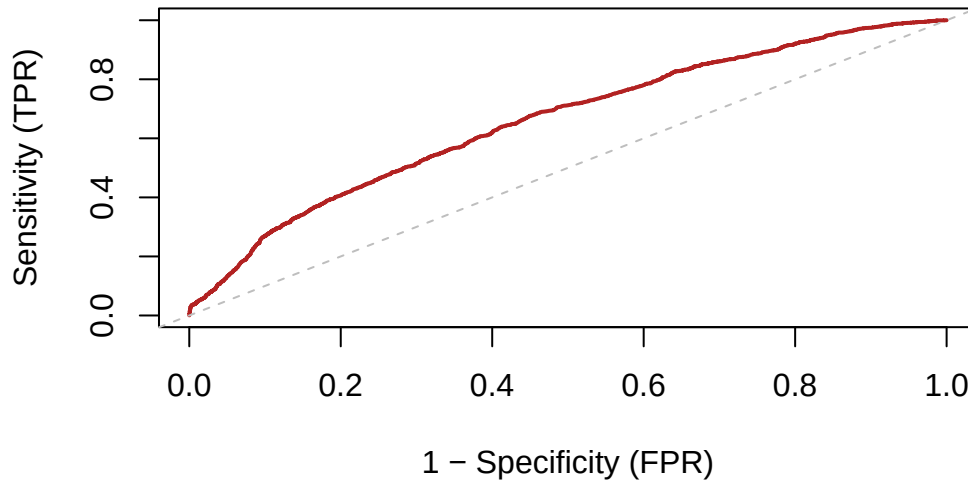
[1] "95% CIs for intercept and sPRS (adjusted model)"

	lower	upper
(Intercept)	0.985025	1.913760
sPRS	2.459660	5.197171

[1] "LRT p-value (adjusted model)"

[1] 4.490765e-98

ROC Curve– sPRS adjusted



```
[1] "AUC for adjusted sPRS log reg"
```

```
[1] 0.6577415
```

Based on these results, we can see that sPRS is significantly associated with metastasis status ($p < 0.01$ in unadjusted and adjusted models). Note the AUCs for both unadjusted and adjusted (0.54, 0.66 respectively) which indicate moderate classification abilities. This is expected from an sPRS. Traditionally, PRSs are not the sole “predictors” for a biological event, and both the p-values and AUCs follow this as sPRS is associated with metastatic status, but is not solely associated with metastasis status. This reflects the biological complexity beyond tumor genetic profiles that influence metastasis.

Section 6: Compare sPRS to similar scores/ predictors

Now that sPRS has been evaluated as a valid “risk score” for metastasis, compare it to other similar scores or predictors, such as TMB, when finding associated metastasis status. Comparisons: TMB, MSI, and Stage Highest Recorded

TMB

Call:

```
glm(formula = y ~ TMB_NONSYNONYMOUS, family = "binomial", data = sPRS_test)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.605302	0.030618	-19.770	< 2e-16 ***
TMB_NONSYNONYMOUS	-0.013029	0.002387	-5.458	4.82e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 8509.1 on 6697 degrees of freedom
Residual deviance: 8466.9 on 6696 degrees of freedom
AIC: 8470.9

Number of Fisher Scoring iterations: 4

[1] "Odds Ratios for TMB"

(Intercept)	TMB_NONSYNONYMOUS
0.5459094	0.9870550

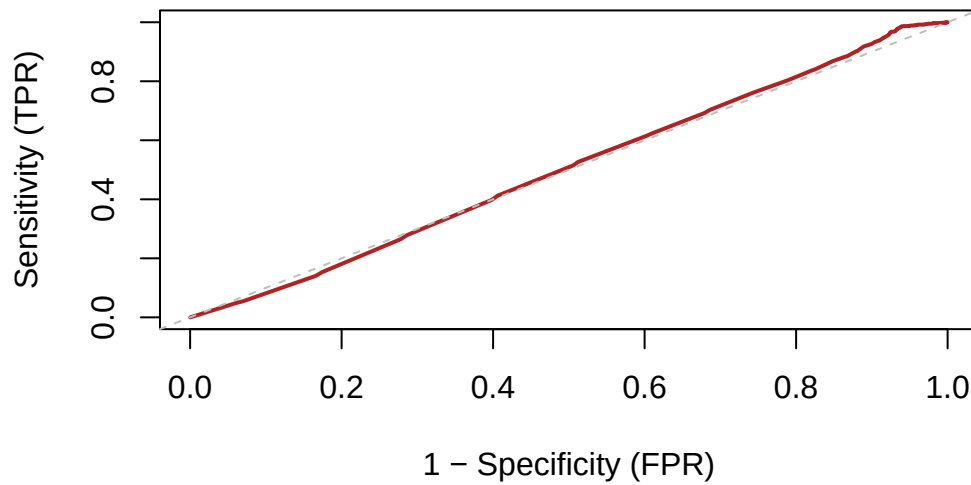
[1] "95% CIs"

	2.5 %	97.5 %
(Intercept)	0.5141483	0.5797196
TMB_NONSYNONYMOUS	0.9822496	0.9914833

[1] "LRT p-value"

[1] 8.287832e-11

ROC Curve- TMB



```
[1] "AUC for TMB"
```

```
[1] 0.5048974
```

The AUC for the TMB model is ~0.50, which is less than the sPRS models (both adjusted and unadjusted). TMB is not a better proxy score for metastasis risk.

MSI

Call:

```
glm(formula = y ~ MSI_SCORE, family = "binomial", data = sPRS_test,  
     na.action = na.omit)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.647020	0.027958	-23.143	< 2e-16 ***
MSI_SCORE	-0.034863	0.007314	-4.767	1.87e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 8501.2 on 6689 degrees of freedom
Residual deviance: 8474.6 on 6688 degrees of freedom
(8 observations deleted due to missingness)
AIC: 8478.6

Number of Fisher Scoring iterations: 4

[1] "Odds Ratios for MSI"

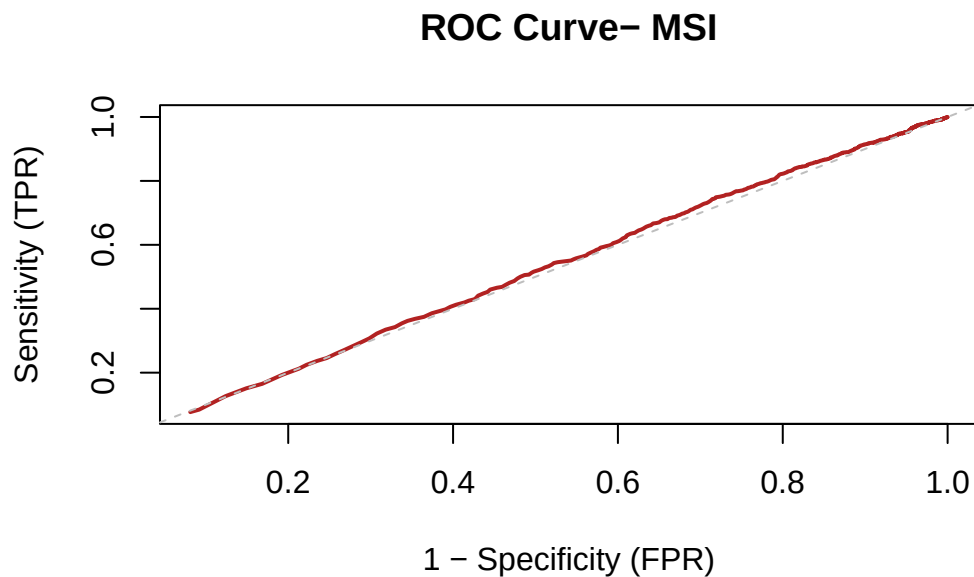
(Intercept)	MSI_SCORE
0.5236038	0.9657374

[1] "95% CIs"

	2.5 %	97.5 %
(Intercept)	0.4956238	0.5530330
MSI_SCORE	0.9515025	0.9792284

[1] "LRT p-value"

[1] 2.447233e-07



```
[1] "AUC for MSI"
```

```
[1] 0.5069402
```

The AUC for the MSI model is ~0.51, which is less than the sPRS models (both adjusted and unadjusted). MSI is not a better proxy score for metastasis risk.

Highest Stage of Cancer Recorded

Call:

```
glm(formula = y ~ as.factor(STAGE_HIGHEST_RECORDED), family = "binomial",  
     data = sPRS_test)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.14540	0.03627	-31.58	<2e-16
as.factor(STAGE_HIGHEST_RECORDED)Stage 4	1.05894	0.05375	19.70	<2e-16

```
(Intercept) ***  
as.factor(STAGE_HIGHEST_RECORDED)Stage 4 ***  
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 8509.1 on 6697 degrees of freedom
Residual deviance: 8113.6 on 6696 degrees of freedom
AIC: 8117.6

Number of Fisher Scoring iterations: 4

```
[1] "Odds Ratios for Stage Highest Recorded"
```

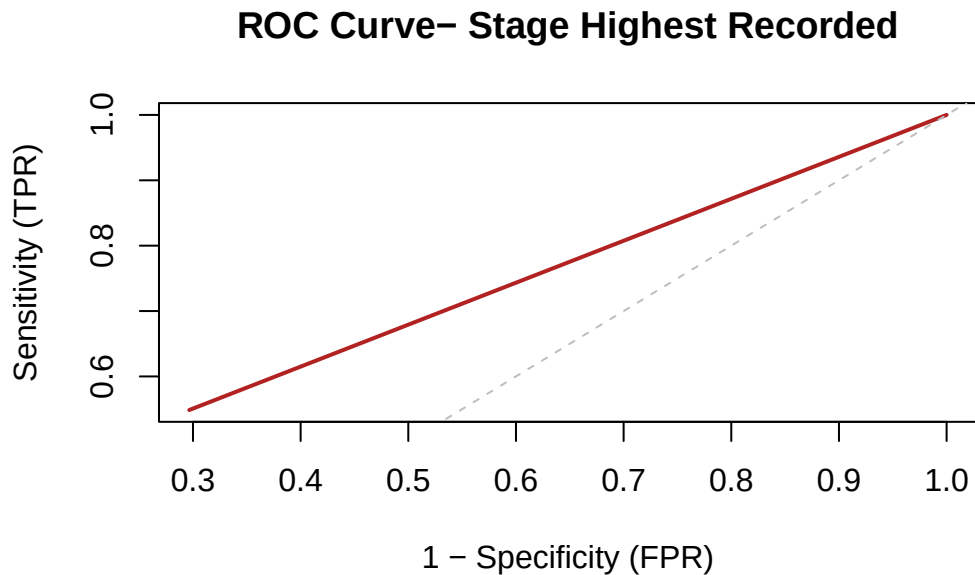
(Intercept)	
	0.3180952
as.factor(STAGE_HIGHEST_RECORDED)Stage 4	
	2.8833147

```
[1] "95% CIs"
```


	2.5 %	97.5 %
(Intercept)	0.2961362	0.3413844
as.factor(STAGE_HIGHEST_RECORDED)Stage 4	2.5954294	3.2042530

[1] "LRT p-value"

[1] 5.329851e-88



[1] "AUC for Stage Highest Recorded"

[1] 0.54469

The highest stage recorded, split into Stages 1-3 and Stage 4, is somewhat better than random guessing to classifying metastasis among cancer patients without adjusting for any other covariate. This is due to the nature of cancer stage classification, where Stage 4 is almost always a metastasized cancer status. This predictor performs similar to sPRS when unadjusted ($\text{auc} = \sim 0.54$), and less than adjusted sPRS ($\text{auc} = \sim 0.66$).

Section 7: Final Tables and Figures

Demographics Table